

## Domácí úloha 1

### Zadání

Pro  $n \in \mathbb{N}$  určete:

$$\begin{pmatrix} -1 & 6 \\ -1 & 4 \end{pmatrix}^n$$

### Řešení

$$A = \begin{pmatrix} -1 & 6 \\ -1 & 4 \end{pmatrix}$$

$$p_A(\lambda) = \begin{vmatrix} -1 - \lambda & 6 \\ -1 & 4 - \lambda \end{vmatrix} = \lambda^2 - 3\lambda + 2 = (\lambda - 1)(\lambda - 2)$$

$$\lambda_1 = 1$$

$$\lambda_2 = 2$$

$$V_1 = \text{Ker} \begin{pmatrix} -2 & 6 \\ -1 & 3 \end{pmatrix} = \langle (3, 1)^T \rangle$$

$$V_2 = \text{Ker} \begin{pmatrix} -3 & 6 \\ -1 & 2 \end{pmatrix} = \langle (2, 1)^T \rangle$$

$$\begin{aligned} \left( \begin{array}{cc|cc} 3 & 2 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right) &\sim \left( \begin{array}{cc|cc} 3 & 2 & 1 & 0 \\ 3 & 3 & 0 & 3 \end{array} \right) \sim \left( \begin{array}{cc|cc} 3 & 2 & 1 & 0 \\ 0 & 1 & -1 & 3 \end{array} \right) \\ &\sim \left( \begin{array}{cc|cc} 3 & 0 & 3 & -6 \\ 0 & 1 & -1 & 3 \end{array} \right) \sim \left( \begin{array}{cc|cc} 1 & 0 & 1 & -2 \\ 0 & 1 & -1 & 3 \end{array} \right) \end{aligned}$$

$$A = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \quad \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix}$$

## Výsledek

$$\begin{pmatrix} -1 & 6 \\ -1 & 4 \end{pmatrix}^n = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2^n \end{pmatrix} \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix}$$

## Domácí úloha 2

### Zadání

Diagonalizujte matici:

$$A = \begin{pmatrix} 2 & 3 & 3 \\ 2 & 1 & 2 \\ -4 & -4 & -5 \end{pmatrix}$$

### Řešení

$$p_A(\lambda) = \begin{vmatrix} 2-\lambda & 3 & 3 \\ 2 & 1-\lambda & 2 \\ -4 & -4 & -5-\lambda \end{vmatrix} = (-\lambda)^3 + \text{Tr} A (-\lambda)^2 + \text{Tr}(\text{adj} A)(-\lambda) + \det A$$

$$\text{adj} A = \begin{pmatrix} 3 & * & * \\ * & 2 & * \\ * & * & -4 \end{pmatrix}$$

$$\det A = -10 - 24 - 24 + 12 + 16 + 30 = 0$$

$$p_A(\lambda) = -\lambda^3 - 2\lambda^2 - 1\lambda$$

$$\lambda^3 + 2\lambda^2 + \lambda = 0$$

$$\lambda(\lambda + 1)^2 = 0$$

$$\lambda_1 = 0$$

$$\lambda_2 = \lambda_3 = -1$$

$$\begin{aligned}
 V_0 &= \text{Ker} \begin{pmatrix} 2 & 3 & 3 \\ 2 & 1 & 2 \\ -4 & -4 & -5 \end{pmatrix} \begin{array}{c} \leftarrow^+ \\ \leftarrow^2 \\ \leftarrow^+ \end{array} \begin{array}{c} \\ -1 \\ \end{array} = \text{Ker} \begin{pmatrix} 0 & 2 & 1 \\ 2 & 1 & 2 \\ 0 & -2 & -1 \end{pmatrix} \begin{array}{c} \leftarrow^1 \\ \leftarrow \\ \leftarrow^+ \end{array} \\
 &= \text{Ker} \begin{pmatrix} 2 & 1 & 2 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \langle (3, 2, -4)^T \rangle
 \end{aligned}$$

$$\begin{aligned}
 V_{-1} &= \text{Ker} \left( \begin{array}{ccc|c} 3 & 3 & 3 & \cdot \frac{1}{3} \left[ \begin{array}{c} \leftarrow^{-1} \end{array} \right]_+^1 \\ 2 & 2 & 2 & \cdot \frac{1}{2} \leftarrow_+ \\ -4 & -4 & -4 & \cdot -\frac{1}{4} \leftarrow_+ \end{array} \right) \\
 &= \text{Ker} \left( \begin{array}{ccc} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right) = \langle (1, 0, -1)^T, (1, -1, 0)^T \rangle
 \end{aligned}$$

$$\begin{aligned} & \left( \begin{array}{ccc|ccc} 3 & 1 & 1 & 1 & 0 & 0 \\ 2 & 0 & -1 & 0 & 1 & 0 \\ -4 & -1 & 0 & 0 & 0 & 1 \end{array} \right) \sim \left( \begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & -1 & 0 \\ 2 & 0 & -1 & 0 & 1 & 0 \\ 0 & -1 & -2 & 0 & 2 & 1 \end{array} \right) \\ & \sim \left( \begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & -1 & 0 \\ 0 & -2 & -5 & -2 & 3 & 0 \\ 0 & -1 & -2 & 0 & 2 & 1 \end{array} \right) \sim \left( \begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & -1 & 0 \\ 0 & 0 & 1 & 2 & 1 & 2 \\ 0 & 1 & 2 & 0 & -2 & -1 \end{array} \right) \\ & \sim \left( \begin{array}{ccc|ccc} 1 & 1 & 0 & -3 & -3 & -4 \\ 0 & 1 & 0 & -4 & -4 & -5 \\ 0 & 0 & 1 & 2 & 1 & 2 \end{array} \right) \sim \left( \begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & -4 & -4 & -5 \\ 0 & 0 & 1 & 2 & 1 & 2 \end{array} \right) \end{aligned}$$

## Výsledek

$$\begin{pmatrix} 2 & 3 & 3 \\ 2 & 1 & 2 \\ -4 & -4 & -5 \end{pmatrix} = \begin{pmatrix} 3 & 1 & 1 \\ 2 & 0 & -1 \\ -4 & -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ -4 & -4 & -5 \\ 2 & 1 & 2 \end{pmatrix}$$

## Domácí úloha 3

### Zadání

Uvažujme matici  $A$ . Najděte všechny matice  $X$  takové, že  $AX$  je násobkem matice  $X$ :

$$A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

### Řešení

Necht':

$$X = x_1 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + x_2 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + x_3 \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + x_4 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$[X]^B = (x_1, x_2, x_3, x_4)^T$$

a dále necht':

$$f(X) = AX,$$

pak:

$$[f]_B^B = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}$$

Pak hledáme taková  $X$ , aby platilo  $f(X) = \lambda X$ , tedy:

$$[X]^B \in \text{Ker} [(AX - \lambda E)]^B$$

$$\begin{aligned} p_f(\lambda) &= \begin{vmatrix} 1-\lambda & 0 & -1 & 0 \\ 0 & 1-\lambda & 0 & -1 \\ -1 & 0 & 1-\lambda & 0 \\ 0 & -1 & 0 & 1-\lambda \end{vmatrix} = \\ &= (1-\lambda) \begin{vmatrix} 1-\lambda & 0 & -1 \\ 0 & 1-\lambda & 0 \\ -1 & 0 & 1-\lambda \end{vmatrix} - 1 \begin{vmatrix} 0 & -1 & 0 \\ 1-\lambda & 0 & -1 \\ -1 & 0 & 1-\lambda \end{vmatrix} \\ &= (1-\lambda)^2 \begin{vmatrix} 1-\lambda & -1 \\ -1 & 1-\lambda \end{vmatrix} - 1 \begin{vmatrix} 1-\lambda & -1 \\ -1 & 1-\lambda \end{vmatrix} \\ &= (\lambda^2 - 2\lambda)(\lambda^2 - 2\lambda) = \lambda^2(\lambda - 2)^2 \end{aligned}$$

$$\begin{aligned} \lambda_1 &= 0 \\ \lambda_2 &= 2 \end{aligned}$$

$$V_0 = \text{Ker} \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix} = \langle (1, 0, 1, 0)^T, (0, 1, 0, 1)^T \rangle$$

$$V_2 = \text{Ker} \begin{pmatrix} -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 \\ -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 \end{pmatrix} = \langle (1, 0, -1, 0)^T, (0, 1, 0, -1)^T \rangle$$

## Výsledek

$$X_0 \in \left\langle \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \right\rangle \quad AX_0 = 0$$

$$X_2 \in \left\langle \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix} \right\rangle \quad AX_2 = 2X_2$$

## Domácí úloha 4

### Zadání

Ukažte, že charakteristický polynom matice  $A$  je až na násobek roven  $\lambda^n + a_{n-1}\lambda^{n-1} + \dots + a_1\lambda + a_0$ , a tedy že ke každému polynomu  $p(\lambda)$  existuje matice, jejímž charakteristickým polynomem  $p(\lambda)$  je:

$$A = \begin{pmatrix} -a_{n-1} & 1 & 0 & 0 & \dots & 0 \\ -a_{n-2} & 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & & \vdots \\ -a_2 & 0 & \dots & 0 & 1 & 0 \\ -a_1 & 0 & \dots & 0 & 0 & 1 \\ -a_0 & 0 & \dots & 0 & 0 & 0 \end{pmatrix}$$

### Řešení

$$p_A(\lambda) = \begin{vmatrix} -a_{n-1} - \lambda & 1 & 0 & 0 & \dots & 0 \\ -a_{n-2} & -\lambda & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & & \vdots \\ -a_2 & 0 & \dots & -\lambda & 1 & 0 \\ -a_1 & 0 & \dots & 0 & -\lambda & 1 \\ -a_0 & 0 & \dots & 0 & 0 & -\lambda \end{vmatrix}$$

$$= (-a_{n-1} - \lambda) \begin{vmatrix} -\lambda & 1 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & & \vdots \\ 0 & \dots & -\lambda & 1 & 0 \\ 0 & \dots & 0 & -\lambda & 1 \\ 0 & \dots & 0 & 0 & -\lambda \end{vmatrix} + (-1) \begin{vmatrix} -a_{n-2} & 1 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & & \vdots \\ -a_2 & \dots & -\lambda & 1 & 0 \\ -a_1 & \dots & 0 & -\lambda & 1 \\ -a_0 & \dots & 0 & 0 & -\lambda \end{vmatrix}$$

$$\begin{vmatrix} -\lambda & 1 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & & \vdots \\ 0 & \dots & -\lambda & 1 & 0 \\ 0 & \dots & 0 & -\lambda & 1 \\ 0 & \dots & 0 & 0 & -\lambda \end{vmatrix} = \begin{vmatrix} -\lambda & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & & \vdots \\ 0 & \dots & -\lambda & 0 & 0 \\ 0 & \dots & 0 & -\lambda & 0 \\ 0 & \dots & 0 & 0 & -\lambda \end{vmatrix} = (-\lambda)^{n-1}$$

$$p_A(\lambda) = (-\lambda)^n - a_{n-1}(-\lambda)^{n-1} + (-1)(-a_{n-2})(-\lambda)^{n-2} +$$

$$+ (-1)^2 \begin{vmatrix} -a_{n-3} & 1 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & & \vdots \\ -a_2 & \dots & -\lambda & 1 & 0 \\ -a_1 & \dots & 0 & -\lambda & 1 \\ -a_0 & \dots & 0 & 0 & -\lambda \end{vmatrix}$$

$$= (-\lambda)^n - a_{n-1}(-\lambda)^{n-1} + (-1)(-a_{n-2})(-\lambda)^{n-2} + \dots +$$

$$+ (-1)^{(n-3)}(-a_2)(-\lambda)^2 + (-1)^{(n-2)} \begin{vmatrix} -a_1 & 1 \\ -a_0 & -\lambda \end{vmatrix}$$

$$= (-\lambda)^n - a_{n-1}(-\lambda)^{n-1} + (-1)(-a_{n-2})(-\lambda)^{n-2} + \dots +$$

$$+ (-1)^{(n-3)}(-a_2)(-\lambda)^2 + (-1)^{n-2}(-a_1)(-\lambda) + (-1)^{(n-1)}(-a_0)$$

$$= (-\lambda)^n - a_{n-1}(-\lambda)^{n-1} + \dots + (-1)^n a_1 \lambda + (-1)^n a_0$$

## Výsledek

$$\forall n = 2k, k \in \mathbb{N} \quad \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0 = p_A(\lambda)$$

$$\forall n = 2k - 1, k \in \mathbb{N} \quad \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0 = -p_A(\lambda)$$

## Domácí úloha 5

### Zadání

Reálná  $5 \times 5$  matice  $A$  má spektrum  $\sigma(A) = \{1, 2, 3, 4, 5\}$ . Určete spektrum matice  $\text{adj}(A)$ .

### Řešení

Pakliže má matice  $A$  spektrum  $\sigma(A) = \{1, 2, 3, 4, 5\}$ , bude její diagonální matice  $D$  vypadat následovně:

$$D = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 5 \end{pmatrix}$$

Determinant  $A$  pak musí být:

$$\det(A) = 120$$

Pak platí:

$$A = RDR^{-1}$$

$$E = A^{-1}RDR^{-1}$$

$$RD^{-1} = A^{-1}R$$

$$D^{-1} = R^{-1}A^{-1}R$$

$$D^{-1} = R^{-1}(\det(A))^{-1}\text{adj}(A)R$$

$$\det(A)D^{-1} = R^{-1}\text{adj}(A)R$$



Přičemž matice  $R^{-1}\text{adj}(A)R$  představuje diagonální matici  $D_{\text{adj}(A)}$  k matici  $\text{adj}(A)$ , na jejíž diagonále se nachází její vlastní čísla, a tedy prvky jejího spektra.

$$D_{\text{adj}(A)} = \det(A)D^{-1} = 120 \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{3} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{5} \end{pmatrix} = \begin{pmatrix} 120 & 0 & 0 & 0 & 0 \\ 0 & 60 & 0 & 0 & 0 \\ 0 & 0 & 40 & 0 & 0 \\ 0 & 0 & 0 & 30 & 0 \\ 0 & 0 & 0 & 0 & 24 \end{pmatrix}$$

## Výsledek

$$\sigma(\text{adj}(A)) = \{120, 60, 40, 30, 24\}$$